

Section 3.1: Integration by Parts

Or The Product Rule for Integrals

Recall: For $y = u(x) \cdot v(x) \longrightarrow y' = u'v + uv'$.

Theorem 11.

$$\int u'(x)v(x) + u(x)v'(x) dx = u(x)v(x) + C$$

$$\text{or } \int u'v + uv' dx = uv + C$$

Theorem 12 (Formula for Integration by Parts). *For u, v differentiable,*

$$\int u dv = uv - \int v du$$

Note 1.

We saw: $u = x^2 \Rightarrow du = 2x dx$

$$u = \quad \Leftarrow du = 3x^2 dx$$

$$v = \quad \Leftarrow du = \cos(x) dx$$

Consider the integrals:

$$\int xe^{x^2} dx, \quad \int x^2 e^x dx, \quad \int xe^x dx, \quad \int e^{x^2} dx$$

Formulae:

$$\int u \, dv = uv - \int v \, du$$

$$\begin{aligned}\int_a^b u \, dv &= \left(uv \, \Big|_{x=a}^{x=b} \right) - \int_a^b v \, du \\ &= \left(u(b)v(b) - u(a)v(a) \right) - \int_a^b v \, du\end{aligned}$$

Example 61. Integrate by parts

1. $\int x e^x \, dx$

2. $\int \ln(x) \, dx$

3. $\int x^2 \ln(x) \, dx$

Example 62. Integrate by parts

1. $\int x^2 e^x dx =$

2. With bounds: $\int_0^1 x^2 e^x dx =$

3. $\int e^x \sin(x) dx =$

Evaluate the integral below. Be sure to place the argument of any trigonometric or logarithmic functions in parentheses in your answer.

1. $\int \sin(\ln(5x)) \, dx =$

2. With bounds:

$$\int_1^5 \sin(\ln(5x)) \, dx =$$

Section 3.2 Trigonometric Integrals

Theorem 13.

$$\sin^2(x) + \cos^2(x) = 1 \qquad \tan^2(x) + 1 = \sec^2(x)$$

$$1 + \cot^2(x) = \csc^2(x)$$

Example 63. Solve

1. $\int \cos^5(x) \, dx$

2. $\int \sin^8(x) \cos^3(x) \, dx$

3. $\int \sin^3(x) \cos^4(x) \, dx$

Theorem 14 (Power-Reducing Formulae).

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \qquad \cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

Example 64. Solve

1. $\int \sin^2(x) \, dx$

2. $\int \cos^4(x) \, dx$

3. Consider: $\int \cos^2(x) \sin(2x) \, dx$

Note: Strategy applies with hyperbolic trig as well: $\cosh^2(x) - \sinh^2(x) = 1$

Section 3.3: Trigonometric Substitution

Consider solving

$$\int \frac{1}{\sqrt{1-x^2}} dx, \quad \int \frac{x}{\sqrt{1-x^2}} dx, \quad \int \frac{1}{x^2\sqrt{1-x^2}} dx$$

Algebra Review: Factoring in and out of parentheses.

Example 65. Factor out the coefficient of x or x^2 .

1. $a^2 + 3ax$

2. $\sqrt{8 - 4x^2}$

3. $\sqrt{4 + 8x^2}$

4. $\sqrt{5 - 17x^2}$

5. $\sqrt{23x^2 + 17}$

Section 3.3: Trigonometric Substitution, continued

Theorem 15. For $a > 0$,

1. for $\sqrt{a^2 - x^2}$, use $x = a \sin(\theta)$ because. . .

$$\sqrt{a^2 - [a \sin(\theta)]^2} =$$

2. for $x^2 + a^2$, try $x = a \tan(\theta)$ because. . .

$$\begin{aligned} [a \tan(\theta)]^2 + a^2 &= a^2 [\tan^2(\theta) + 1] \\ &= a^2 \sec^2(\theta) \end{aligned}$$

3. for $\sqrt{x^2 - a^2}$, try $x = a \sec(\theta)$ because. . .

$$\begin{aligned} \sqrt{[a \sec(\theta)]^2 - a^2} &= \sqrt{a^2 [\sec^2(\theta) - 1]} \\ &= \sqrt{a^2} \cdot \sqrt{[\sec^2(\theta) - 1]} \\ &= a \tan(\theta) \end{aligned}$$

Example 66. Solve using trigonometric substitution.

1. $\int \frac{1}{(\sqrt{5 - x^2})^3} dx =$

Example 67. Solve using trigonometric substitution. (Use an integral table for the θ integral if needed.)

2. $\int \frac{1}{x\sqrt{11-x^2}} dx =$

3. Prove the identity with trig sub: $\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$

Important: Might need **double-angle formula** for sine or cosine. (Not needed on Mini-Exams.)

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$$

$$\cos(2\theta) = 1 - 2 \sin^2(\theta)$$

$$\sin(4\theta) = 2 \sin(2\theta) \cos(2\theta) = 2 \cdot 2 \cdot \sin(\theta) \cos(\theta) (1 - 2 \sin^2(\theta))$$

Example.

$$\begin{aligned} \frac{225}{8} \int (1 - \cos(4\theta)) d\theta &= \frac{225}{8} \left(\theta - \frac{1}{4} \sin(4\theta) \right) + C \\ &= \frac{225}{8} \theta - \frac{225}{32} \sin(4\theta) + C \\ &= \frac{225}{8} \theta - \frac{225}{16} \sin(2\theta) \cos(2\theta) + C \\ &= \frac{225}{8} \theta - \frac{225}{8} \sin(\theta) \cos(\theta) (1 - 2 \sin^2(\theta)) + C \end{aligned}$$

Example. Discussion about bounds:

Step 1. First, rewrite the integral so that we can easily identify a .

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{36x^2 + 144}} dx &= \int_0^1 \frac{1}{\sqrt{6^2 (2^2 + x^2)}} dx \\ &= \frac{1}{6} \int_0^1 \frac{1}{\sqrt{2^2 + x^2}} dx \end{aligned}$$

Looking at the rewritten integral, we see that $a = 2$. Thus, we will use the substitution $x = 2 \tan \theta$ and $dx = 2 \sec^2 \theta d\theta$. We must change the limits of integration. If $x = 0$, then $0 = 2 \tan \theta$ and, hence, $\theta = 0$. If $x = 1$, then $1 = 2 \tan \theta$ and, hence, $\theta = \tan^{-1}(\frac{1}{2})$. After making these substitutions, we have

$$\frac{1}{6} \int_0^1 \frac{1}{\sqrt{2^2 + x^2}} dx = \frac{1}{6} \int_0^{\tan^{-1}(\frac{1}{2})} \frac{1}{\sqrt{2^2 + 2^2 \tan^2 \theta}} 2 \sec^2 \theta d\theta.$$

Step 2. Simplifying the integral, we obtain

$$\begin{aligned} \frac{1}{6} \int_0^{\tan^{-1}(\frac{1}{2})} \frac{1}{\sqrt{2^2 + 2^2 \tan^2 \theta}} 2 \sec^2 \theta d\theta &= \frac{1}{6} \int_0^{\tan^{-1}(\frac{1}{2})} \frac{2 \sec^2 \theta}{2 \sec \theta} d\theta \\ &= \frac{1}{6} \int_0^{\tan^{-1}(\frac{1}{2})} \sec \theta d\theta. \end{aligned}$$

Advanced Examples... (Fully Optional)

1. $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$

2. $\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + c$

3. Completing the square: $x^2 + bx + c = (x - h)^2 + k$, for $h = -\frac{b}{2}$, $k = c - \frac{b^2}{4}$.

Thus

$$\int \frac{1}{x^2 + bx + c} dx = \int \frac{1}{k + (x - h)^2} dx$$

=

Section 3.3: Trigonometric Substitution, continued

Example 68. Evaluate the integrals

$$\int \frac{1}{x^2 + 6x + 13} dx =$$

$$\int \frac{1}{4x^2 - 8x + 29} dx =$$

Section 3.4: Partial Fractions

Notice:

$$\text{Poly Division: } \frac{N(x)}{D(x)} \iff \text{Partial Frac. Decomp.: } \frac{D(x)}{N(x)}$$

Example 69. (Optional Review.) Evaluate the following by first simplifying with polynomial division.

1. $\int \frac{x^3 + 1}{x} dx =$

2. $\int \frac{3x^3 + x^2 - 14x - 5}{x^2 - 5} dx =$

Expanding with Partial Fractions

Patterns: For $\deg(\text{numerator}) < \deg(\text{denominator})$, there are numbers $A_1, A_2, A_3, \dots$

$$1. \frac{\text{numerator}}{(x-c)^2} = \frac{A_1}{x-c} + \frac{A_2}{(x-c)^2}$$

$$2. \frac{\text{numerator}}{(x-c)^3} = \frac{A_1}{x-c} + \frac{A_2}{(x-c)^2} + \frac{A_3}{(x-c)^3}$$

$$3. \frac{\text{numerator}}{(x-c)^4} = \frac{A_1}{x-c} + \frac{A_2}{(x-c)^2} + \frac{A_3}{(x-c)^3} + \frac{A_4}{(x-c)^4}$$

$$4. \frac{\text{numerator}}{(x-b)^2(x-c)^4} = \frac{A_1}{x-b} + \frac{A_2}{(x-b)^2} + \frac{B_1}{x-c} + \frac{B_2}{(x-c)^2} + \frac{B_3}{(x-c)^3} + \frac{B_4}{(x-c)^4}$$

And $B_1, B_2, B_3, \dots$ and $C_1, C_2, C_3, \dots$

$$1. \frac{\text{numerator}}{(x-c)(ax^2+bx+d)} = \frac{A_1}{x-c} + \frac{B_1x+C_1}{ax^2+bx+d}$$

$$2. \frac{\text{numerator}}{(x-c)^2(ax^2+bx+d)^2} = \frac{A_1}{x-c} + \frac{A_2}{(x-c)^2} + \frac{B_1x+C_1}{ax^2+bx+d} + \frac{B_2x+C_2}{(ax^2+bx+d)^2}$$

Key to second kind: irreducibility of: ax^2+bx+d

Example 70. Decompose into partial fractions.

$$1. \frac{5}{(x-2)^2} =$$

2. Expand using PFD. (Do NOT solve for the coefficients.)

$$\frac{1-17x}{x^2(x+27)} =$$

3. $\frac{x+1}{(x-1)^2}$

4. $\frac{2x}{(x-1)(3x+1)}$

$$5. \frac{19 - 2x^2}{(x - 1)^2(3x + 1)} =$$

6. Expand using PFD. (Do NOT solve for the coefficients.)

$$\frac{5x - 7x^4}{(x - 1)^2(3x + 1)^3} =$$

7. Expand using PFD. (Do NOT solve for the coefficients.)

$$\frac{x^2 - 1}{x^3(x^2 + 1)^2} =$$

Example 71. Using the previous example, **discuss** the calculation these integrals.

1. From a PFD calculator: $\frac{2(x+1)}{x^2(x^2+2)} = \frac{1}{x} + \frac{1}{x^2} - \frac{x+1}{x^2+2}$

$$\begin{aligned} \int \frac{2(x+1)}{x^2(x^2+2)} dx &= \\ &= \int \frac{1}{x} dx + \int \frac{1}{x^2} dx - \int \frac{x+1}{x^2+2} dx \\ &= \int \frac{1}{x} dx + \int \frac{1}{x^2} dx - \end{aligned}$$

2. Do the following: (1) Expand using PFD; (2) Evaluate the integral using generic constants in the numerator: A, B, C .

$$\int \frac{2x}{(x-1)^2(3x+1)} dx =$$

Basic Review

Integration Review

Example 72. 1. $\int s(s-1)^9 ds$

2. $\int (t-6)\sqrt{t-3} dt$

3. $\int \frac{x+2}{x^2+4} dx$

4. $\int \frac{1}{x^2-2x+10} dx$

Example 73. Evaluate

1. $\int \frac{x^{5/2} - x^{1/2}}{x^{3/2}} dx$

2. $\int e^x \csc(3e^x + 4) dx$

3. $\int \frac{1-x}{1-\sqrt{x}} dx$

4. $\int \frac{x^2 + 2}{x - 2} dx$